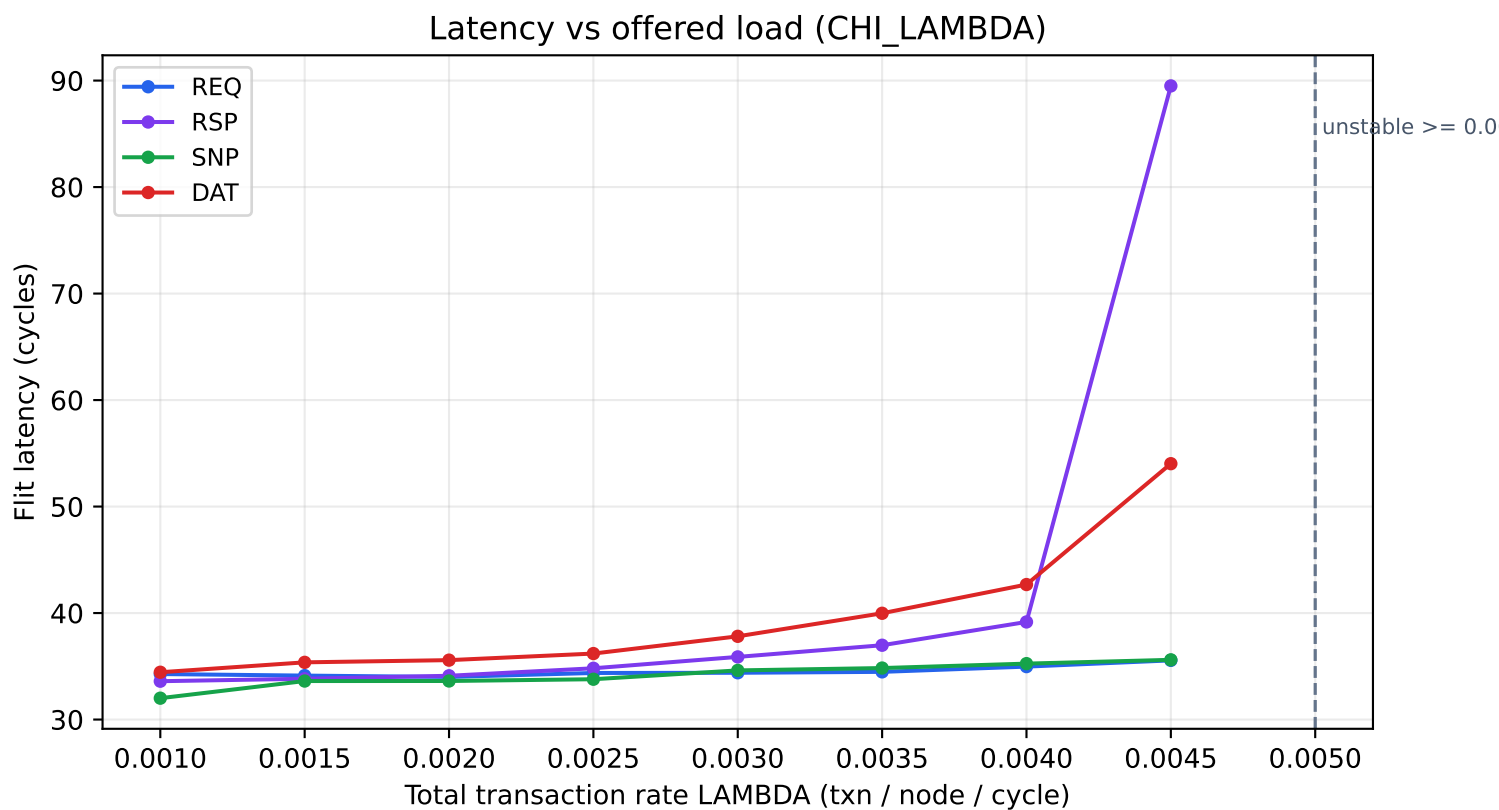
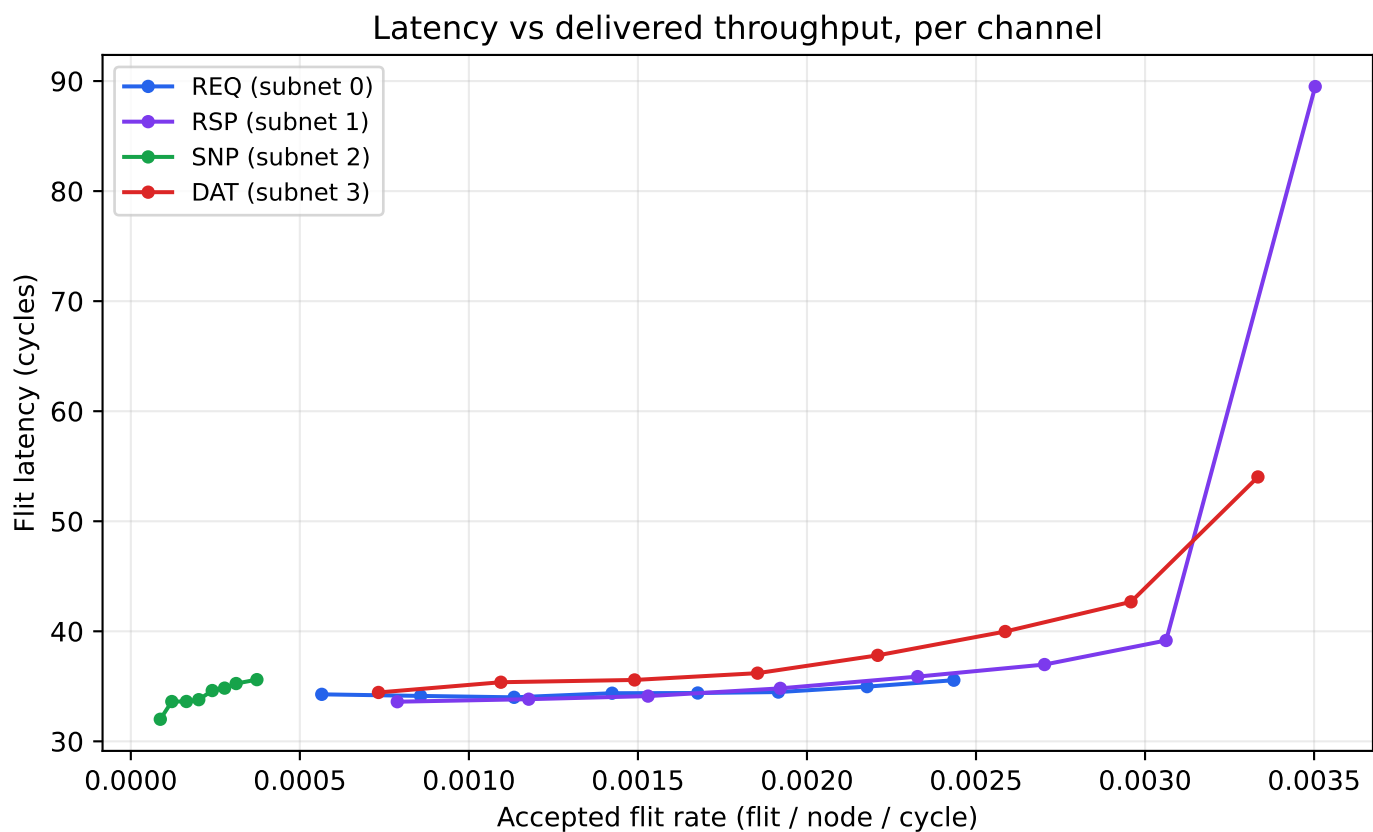
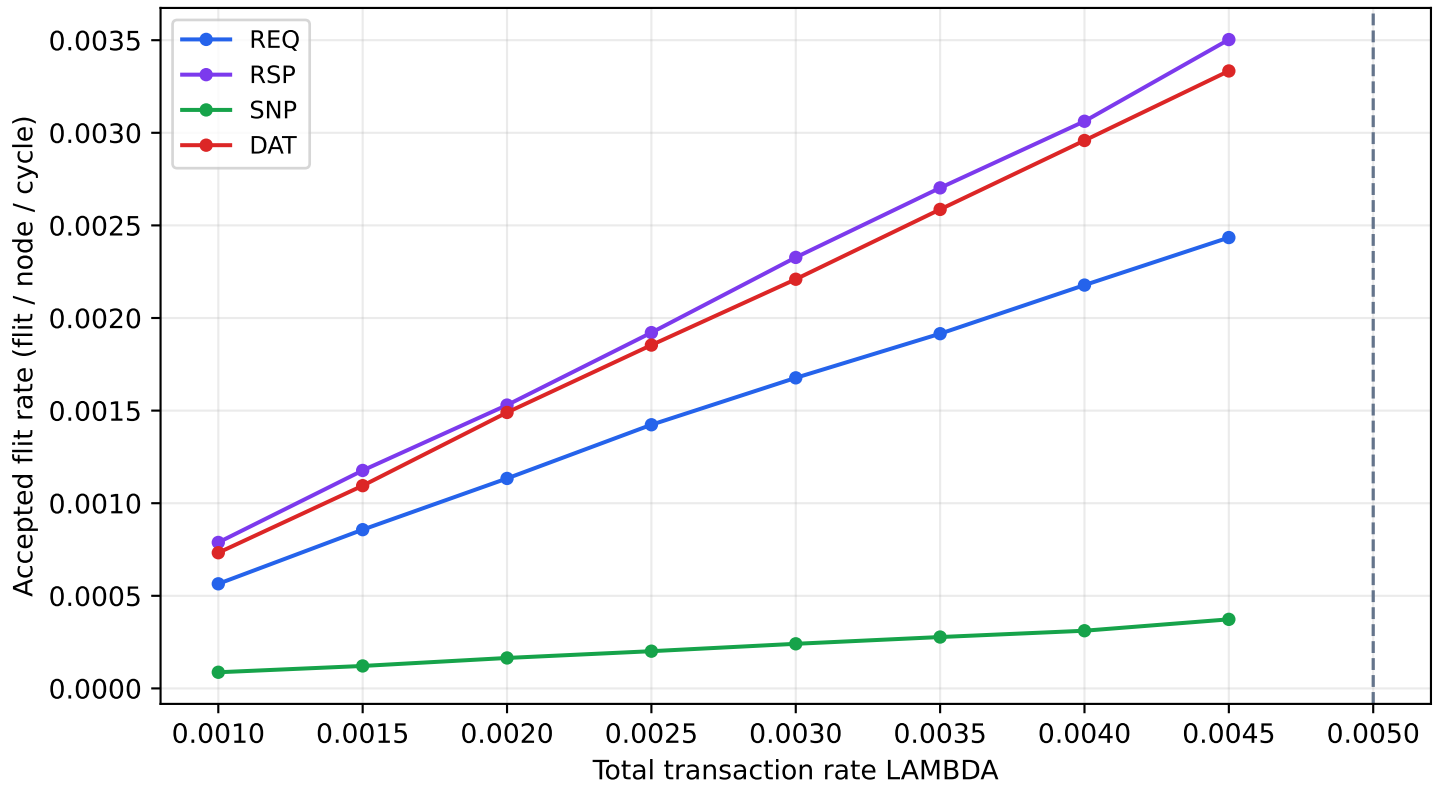


CHI 4-Channel Latency-Throughput Sweep

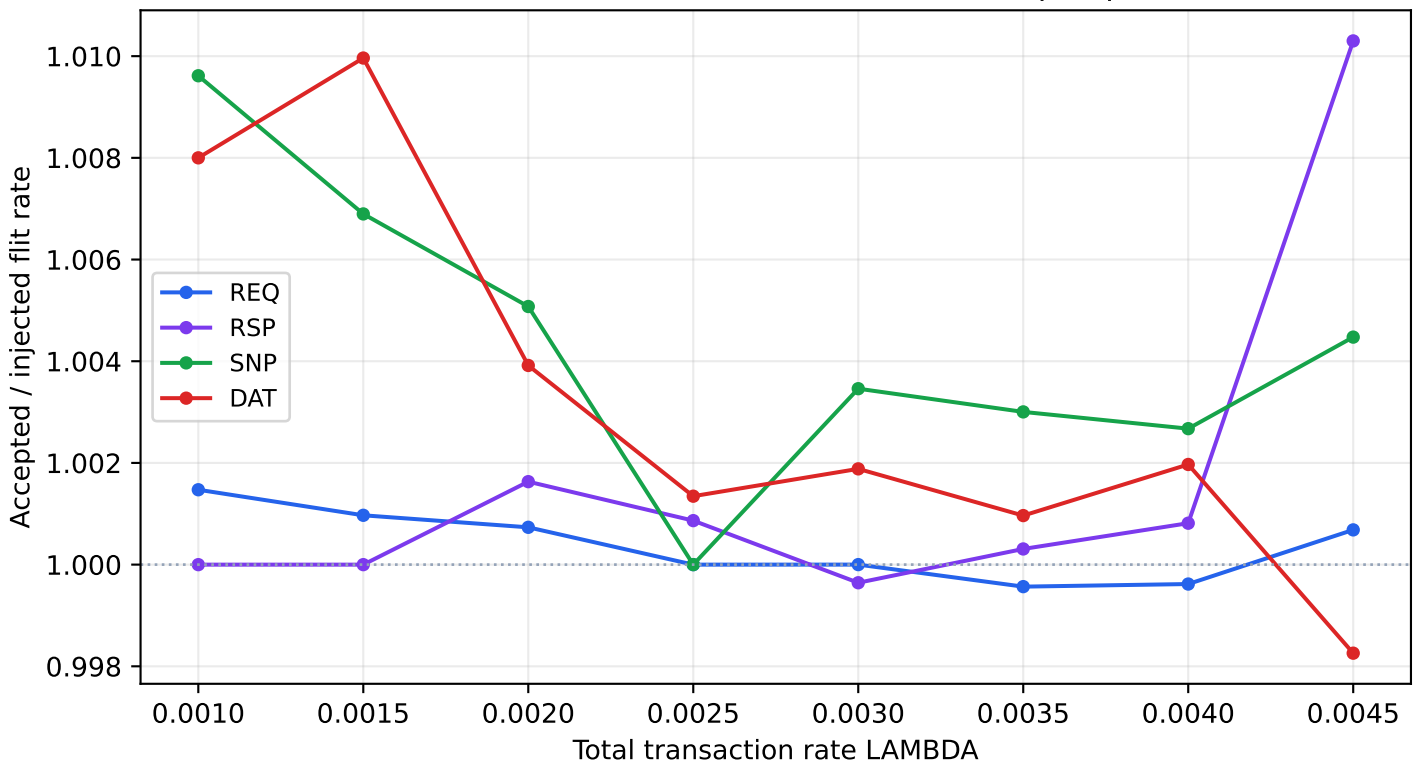


Throughput Scaling & Acceptance

Per-channel accepted throughput vs offered load



Drain ratio (≈ 1 means the channel keeps up)



Saturation Summary & Optimization

Channel	Subnet	Last stable LAMBDA	Accepted flit @ last stable	Flit lat @ last stable	Knee LAMBDA	Knee flit lat
REQ	0	0.0045	0.00243	35.6	0.0045	35.6
RSP	1	0.0045	0.00350	89.5	0.0045	89.5
SNP	2	0.0045	0.00037	35.6	0.0015	33.6
DAT	3	0.0045	0.00333	54.0	0.0045	54.0

- Network saturates near LAMBDA ~ 0.005 txn/node/cycle; the next swept point goes unstable (latency > 500 cycles)
- First channels to bend up are RSP and DAT (subnets 1/3); REQ and SNP stay flat to the knee.
- Saturation accepted throughput is low (~0.003 flit/node/cycle/channel), so the limiter is destination hotspot concentration
- DAT carries 2-flit payloads and concentrates on the 4 SN nodes (DMT reads, L3 evictions, CleanInvalid writeback); th
- RSP has the most classes and fans into RN/HN; protect it with deeper VC buffers or an extra VC rather than treating
- Optimization 1: add more SN/DDR attach points (or spread the 4 SNs across non-adjacent routers) to widen the men
- Optimization 2: raise vc_buf_size (2 -> 4/8) and num_vcs on RSP/DAT subnets; shallow 2-flit buffers cap the knee we
- Optimization 3: place HN home-node interleaving so REQ/RSP do not all converge on a few HN tiles; balance hotspot
- Optimization 4: if DAT stays dominant, widen DAT (more lanes / larger flit) or split read-data vs write-data into separ